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DSC 140S Final Project: Baseball Data Analysis

Introduction:

For my final project I will be looking at a baseball data set of stats from all rookie pitchers from 2024. This data set was chosen to see if looking at rookie pitchers' data can give any insight on how to run or coach a team. First off, let's specify the different statistics that are in this data set. Win and lose percentage, or W-L%, is the pitchers wins divided by the number of wins and losses. A pitcher gets a win in the MLB when a pitcher is still pitching when their team takes the lead in a win. Walks and hits per innings pitched, or WHIP, is a stat that tells on average how many walks or hits a pitcher gives up per inning they pitch. Strike outs, or SO, is the number of strikes outs a pitcher has. Hit by pitches, or HBP, is the number of times a pitcher has hit a batter. Wild pitches, or WP, is the number of wild pitches a pitcher has. Age is the age of the pitcher. Earned run average, or ERA, is the number of runs a pitcher is statistically supposed to give up if they pitched nine innings. Innings pitched, or IP is the number of innings a pitcher has pitched. Wins, or W, is the number of wins a pitcher has. Hits, or H, is the number of hits a pitcher has given up. Years, or Yrs, is the number of years in the majors it took this rookie to complete their rookie year. Sometimes when a new rookie gets called up to the big leagues he will not play enough games or pitch enough innings to count as a full rookie season, so when a player has 2 years it means they played enough for their rookie year to count in the second year of being up with the big league team. The From and To columns represent the year they first got called up to the big and the year that their rookie year counted. All-star games, or ASG, represents if they were an all-star their rookie year. This does not happen too often, although there are a couple pitchers in this data set who did get recognized as an all-star. Wins above replacement, or WAR, represents how much better or worse a pitcher is compared to the average pitcher. If the pitcher is better than the average pitcher, then they will have a positive WAR. Looking at this data set can allow someone to compare different statistics to see if there are any patterns or predictions that can help them decide on who to call up to the big-league team or who to pitch.

Data Set: 2024 Major League Baseball Rookie Players | Baseball-Reference.com

Description of the Data Set:

Before looking at any statistics the data set was cleaned by removing the null values and the totals column. The first way of looking at the data sets specifics was to use the describe function in Python. This allows me to see statistics like the average, standard deviation, and maximum of some of the columns of the data set. Here is an image of the statistics of some of the columns of data from the data set.

	Rk	Yrs	From	To	ASG	WAR	W	L	W-L%	ERA	...	HR	BB
count	178.000000	178.000000	178.000000	178.0	178.000000	178.000000	178.000000	178.000000	178.000000	178.000000	...	178.000000	178.000000
mean	131.522472	1.556180	2023.308989	2024.0	0.016854	0.330337	2.505618	2.853933	0.471157	5.358652	...	6.696629	18.797753
std	79.835509	0.766304	1.073580	0.0	0.129087	0.982376	2.802838	2.812605	0.331686	3.958179	...	6.198969	14.828343
min	3.000000	1.000000	2018.000000	2024.0	0.000000	-1.600000	0.000000	0.000000	0.000000	0.000000	...	0.000000	0.000000
25%	61.500000	1.000000	2023.000000	2024.0	0.000000	-0.300000	1.000000	1.000000	0.250000	3.395000	...	2.000000	8.250000
50%	129.000000	1.000000	2024.000000	2024.0	0.000000	0.100000	1.000000	2.000000	0.477500	4.540000	...	5.000000	15.000000
75%	201.750000	2.000000	2024.000000	2024.0	0.000000	0.700000	3.000000	4.000000	0.667000	6.245000	...	9.000000	25.000000
max	272.000000	5.000000	2024.000000	2024.0	1.000000	5.900000	16.000000	13.000000	1.000000	34.710000	...	27.000000	98.000000

caption: This table shows the count, mean (average), std (standard deviation), min (minimum), 25% (first quartile), 50% (second quartile), 75% (third quartile), and max (maximum) for multiple columns of data.

When looking at this table there are a bunch of statistics that stand out to me. The first one is the maximum ERA compared to the average and the quartiles. A good ERA in the MLB is considered to be between 3.00 and 4.00. The maximum ERA is around 34.7, which means that this pitcher gave up about 34 runs per nine innings pitched. The mean ERA is only about 5.3 though and the quartiles are also significantly lower. This shows that the rookie with the max ERA most likely didn't pitch a lot of innings. If a pitcher doesn't pitch nine innings and only gives up a couple of runs their ERA will be inflated because it will be based off if they were to throw nine innings. You see this a lot with relief pitchers due to them coming out of the bullpen and not getting as many innings as the starters. The next stat that stood out was the maximum for the year's column. The maximum in the year's column is 5, which means he got called up 4 times and didn't play enough games to be considered a rookie. This usually doesn't happen, and most rookies take 1 to 2 years. This could be a possible outlier in the data. The final stat that stood out was the average for the ASG column. The average for all-star games is about 0.02. This stat alone shows how rookies usually don't become all stars in their rookie year.

The next part of statistics will be looking at the possible values for the teams, or Tm, column.

```
array(['CIN', 'LAA', 'NYY', 'HOU', 'SFG', 'MIA', 'OAK', 'TBR', 'CHW',
       'TOT', 'MIN', 'BOS', 'KCR', 'CHC', 'PIT', 'NYM', 'CLE', 'LAD',
       'ARI', 'COL', 'SEA', 'ATL', 'STL', 'TEX', 'MIL', 'DET', 'WSN',
       'PHI', 'SDP', 'TOR', 'BAL'], dtype=object)
```

Caption: This image shows all the possible values in the team's column. Each three-letter acronym stands for a different MLB team.

I can tell from this that there are 31 possible values. This is actually very interesting because there are only 30 teams in the MLB. I looked through the possible values and found TOT. TOT isn't the initials for any MLB team, so I'm guessing it means that that player played for more than one team in their rookie year or over the years until they counted as a rookie. Those are only some of the oddities found in the data set and taking a deeper look could show a lot more.

Data Analysis

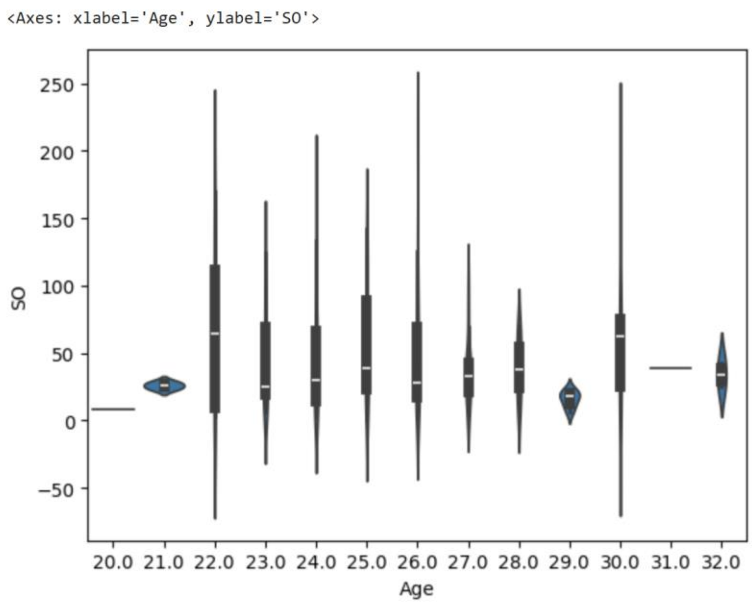
Question 1:

Does age affect the number of strikeouts or walks a rookie pitcher has?

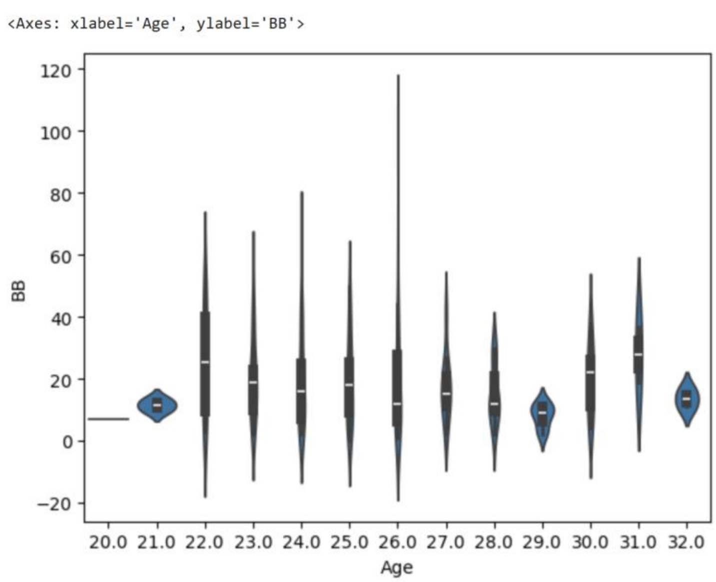
Process:

The process to find how age affects strikeouts and walks is by making two violin plots. Violin plots are another version of a box and whisker plot, but it also shows the distribution of data a little better with the boxes moving in or out from the median. It will also show the range and the quartiles of the data. A violin plot was made for strikeouts and age and walks and age.

Graphs:



caption: This graph shows the distributions of data on a violin graph for strikeouts and age. Each box shows the range and quartiles for each age.



caption: This graph shows the distributions of data on a violin graph for walks and age. Each box shows the range and quartiles for each age.

Analysis:

In the first violin graph it is easy to tell that most of the data points are between the ages of 22 and 26. This makes sense because pitchers are either developed in college or the minors for a couple years. This age range must show that these are the ages you'll get the most strikeouts out of other than the age of 30. Looking at the second graph almost shows the same results with the greatest number of walks on average coming from the age range of 22 to 26. The largest range on this graph is with the age of 26. This makes sense with young pitchers because one of the big parts they struggle with early is command. Also, the ages of 20, 21, 29, 31, and 32 are the ages with the lowest number of strikeouts and walks. Looking at the age ranges showed that the ages with the most strikeouts are the ages most likely to get called up to the majors, while the ones with a lower number of strikeouts don't get called up to the majors at those ages as often and it's about the same with the walks graph. This can help when looking at the ages of possible pitchers to call up, especially if you are looking for them to strike out a lot of batters.

Questions 2:

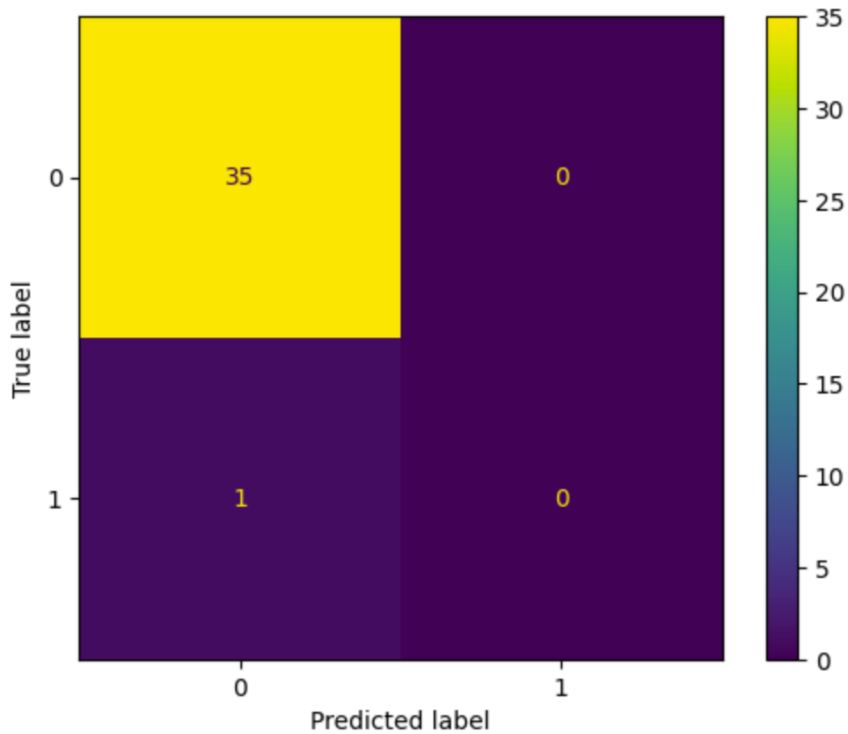
Can you predict if a rookie pitcher will be an all-star?

Process:

To answer this question, complete a train split test with WAR, ERA, and WHIP as the x components and ASG as the y component. The size for the test data is 20%, which is the amount of data that will be used to make predictions, and it is looking at three neighbors, which is the number of surrounding data points that each data point will look at. This will make the predictions then compare them to the actual data in a confusion matrix and it will calculate and print the accuracy score for the predictions.

Results:

```
97.22222222222221
<sklearn.metrics._plot.confusion_matrix.ConfusionMatrixDisplay at 0x2ac276dfcb0>
```



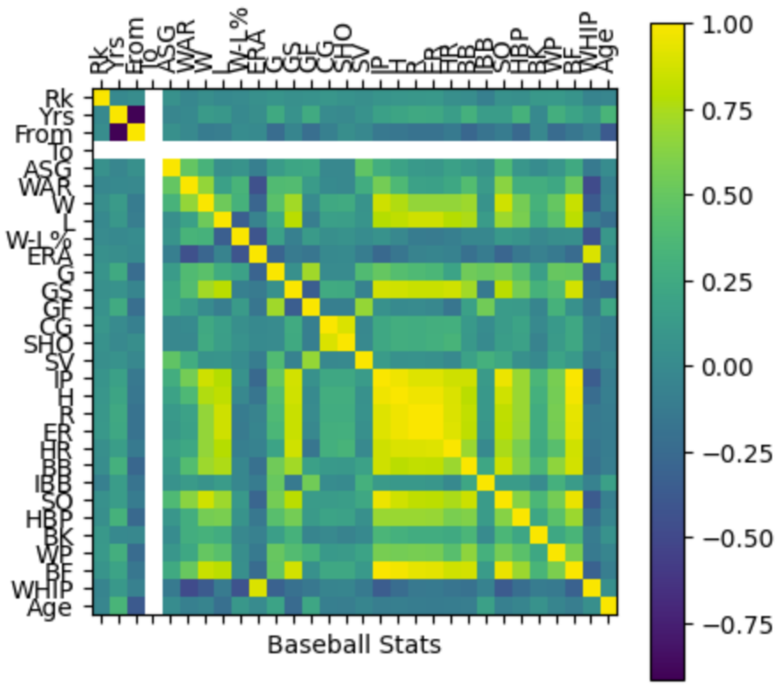
caption: This is a confusion matrix that shows the predictions vs the actual data. It shows the predictions of all-stars based on WAR, ERA, and WHIP. The number of neighbors is 3 and the test size is 20%. The accuracy score is printed in the top left. 0 stands for no all-star appearance and 1 stands for an all-star appearance.

Analysis:

Based on the accuracy score and confusion matrix the training algorithm went well. It got an accuracy score of about 97% which means that the training worked almost perfectly. From the confusion matrix I can tell that the 1 player was an all-star, when looking at the data it predicted that there would be no all-stars. This makes sense based on the fact that not a lot of rookies become all stars in their rookie year. The rookies that do become all stars can be considered outliers because of the low amount of them.

Improvements:

Using feature engineering and hyperparameter tuning to help improve the predictions. For feature engineering create a correlation matrix that will show the columns that most correlate to ASG. Hyperparameter tuning will show the best number of neighbors to use when re-doing the predictions.

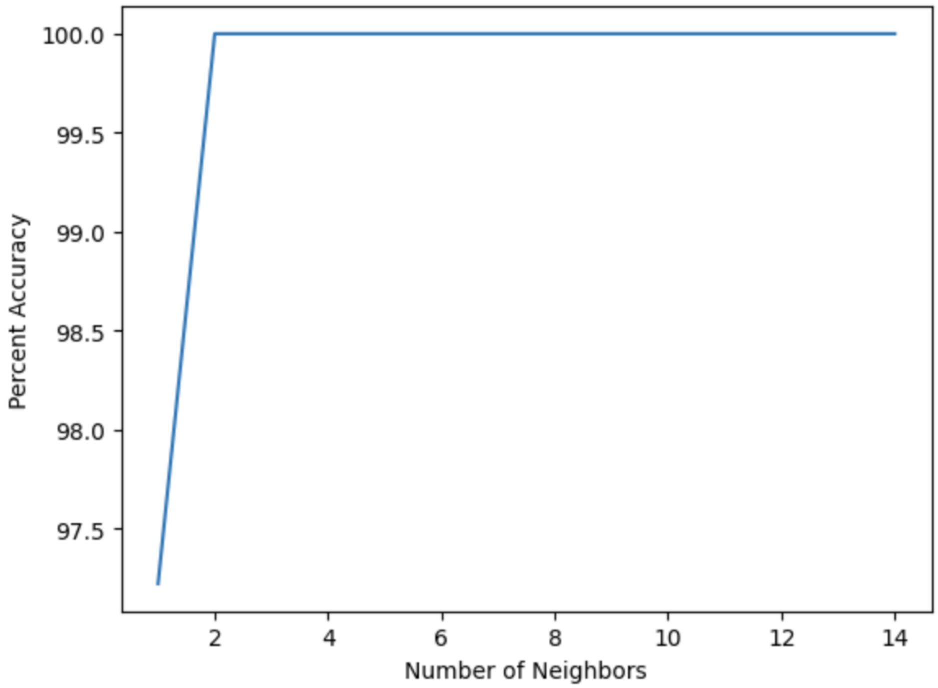


caption: This is a confusion matrix that shows the correlation between all the columns of data. The columns that match up with the colors closer to 1.0 and –1.0 are the columns that most correlate with each other.

Analysis:

I can tell from this correlation matrix that WAR, saves, and strike outs will correlate best with the All-Star Games column.

```
Text(0, 0.5, 'Percent Accuracy')
```



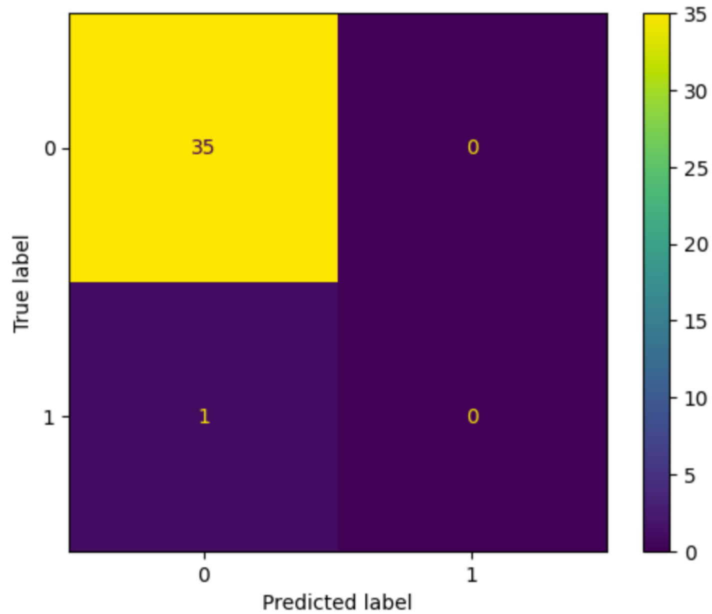
caption: This graph shows the number of neighbors and the percentage of accuracy that will be predicted.

<u>Analysis:</u>

It's easy to tell from this table that the number of neighbors doesn't matter a whole lot for this test. Any number of neighbors between 2-14 will predict at around 100%.

<u>Results:</u>

```
97.22222222222221
<sklearn.metrics._plot.confusion_matrix.ConfusionMatrixDisplay at 0x2ac28671f90>
```



caption: This is a confusion matrix that shows the predictions vs the actual data after the improvements. It shows the predictions of all-stars based on WAR, SV, and SO. The number of neighbors is 3 and the test size is 20%. The accuracy score is printed in the top left. 0 stands for no all-star appearance and 1 stands for an all-star appearance.

After using feature engineering and hyperparameter tuning the accuracy score stayed the same. The only prediction that it got wrong was that it did predict everyone wouldn't be an all-star, but one player was. This is the same incorrect prediction that was made before the improvement. Again, this is one of the easier statistics to predict because of the low number of rookie all-stars. The test could make a 100% prediction test, but it would most likely be if none of the players that were all stars were in the percentage of data that was tested. Overall, the prediction of all-stars as rookies went well, but it is affected by the low amount of rookie all-stars.

Question 3:

Do the strikeouts and walks column rely on each other and what is their relationship?

Process:

Performing a Chi-squared test and finding the spearman correlation score will show if the columns rely on each other and what type of relationship they have. The Chi-squared test will give me a p-value and if the p-value < 0.05 then the columns rely on each other. The spearman correlation score will print a number between 1.0 and –1.0. If the correlation score is closer to 1.0 or –1.0 it means that the columns have a strong relationship with each other. If the correlation score is closer to zero in either direction it means the columns don’t have a strong relationship.

Results:

- Chi-squared test p-value = about 1.21
- Spearman Correlation score = about 0.86

Analysis:

With the p-value being greater than 0.05, this means that the strikeouts and walks columns do not rely on each other. Although, when looking at the spearman correlation score at 0.86, it is close to 1.0 so you can see that the two columns have a strong positive relationship. This means that as the number of strikeouts goes up the number of walks does as well. This could come from pitchers throwing more innings resulting in more opportunities for a strikeout or walk. Also, this brings us back to the point before that rookie pitchers struggle with command because even if you have more strikeouts that just means the pitcher is also likely to have more walks. This can help let a team know that if you have a rookie pitcher that strikes out a lot of batters, then you should expect him to walk a decent number of hitters as well.

Conclusion:

After doing this research you can tell that doing tests and comparing other statistics can help you when looking at the ability of different pitchers and what to expect out of them. Having this knowledge can help coaches and organizations decide who to pitch, when to pitch a rookie, and what pitchers to call up to the majors.

Works Cited

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